IoT Based Automatic Solar Panel Cleaning System

Batch No:A4

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Introduction

- Solar energy is one of the fastest-growing renewable energy sources worldwide
- It provides a clean, sustainable, and environmentally friendly alternative to fossil fuels.
- Solar Photovoltaic (PV) panels convert sunlight directly into electricity.
- Governments and industries are investing heavily in solar installations to reduce carbon emissions.
- Solar systems are widely deployed in residential, commercial, industrial, and utility-scale applications.

OBJECTIVES

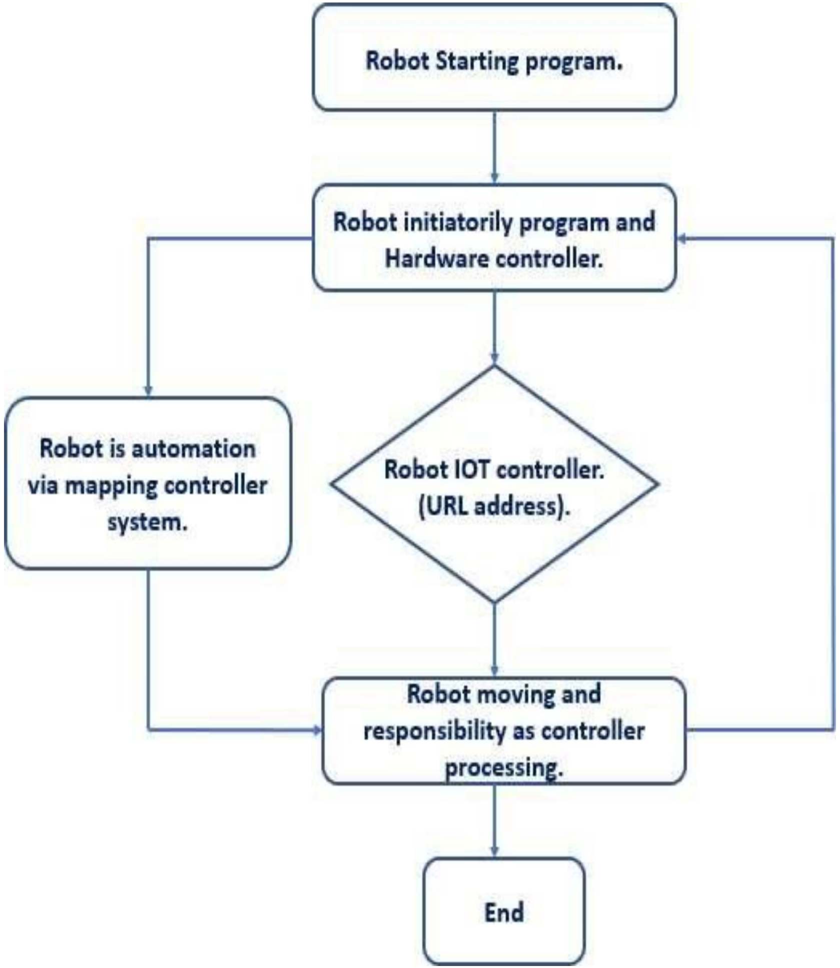
- Detect light intensity using LDR sensor
- Monito solar panel performance
- Process data using ESP32 controller
- Operate cleaning motor using motor driver
- Automate solar panel cleaning system
- Enable WiFi-based monitoring/control

Problem Statement:

Solar panels rely on direct sunlight to generate electricity efficiently. However, dust, dirt, and environmental pollutants accumulate on the panel surface over time, blocking sunlight from reaching the photovoltaic (PV) cells. This obstruction can reduce energy generation efficiency by up to 15–30%, and in heavily polluted or dusty regions, the loss can be even higher. Although manual cleaning is commonly used to restore performance, it is time-consuming, requires significant manpower, wastes large amounts of water, and is not practical for large-scale solar farms or rooftop installations. Therefore, an automated and intelligent maintenance solution is essential to ensure consistent solar panel performance and maximize energy output.

Block Diagram

Solar Panel
↓
LDR Dust Sensor
↓
NodeMCU (ESP8266 Controller)
↓
Motor Driver
↓
DC Motor / Pump
↓
Brush / Wiper Cleaning System
↓
WiFi / Internet
↓
IoT Cloud (Blynk)
↓
Mobile App Monitoring



Proposed Solution

- IoT-based automatic cleaning system.
- Sensors detect dust accumulation.
- Microcontroller activates motorized brush.
- Real-time monitoring via mobile app.
- Reduces maintenance cost and human effort.

List of main components

1. NodeMCU (ESP8266) or ESP32 Microcontroller
2. LDR Sensor (Dust / Light Detection)
3. Motor Driver Module (L298N / L293D)
4. DC Motor or Water Pump (Cleaning Mechanism)
5. Cleaning Brush or Wiper
6. Power Supply (Battery or Adapter)
7. Arduino IDE (Programming Software)
8. IoT Platform (Blynk / ThingSpeak / MQTT)
9. Mobile App / Dashboard (Monitoring & Control)

Hardware Components

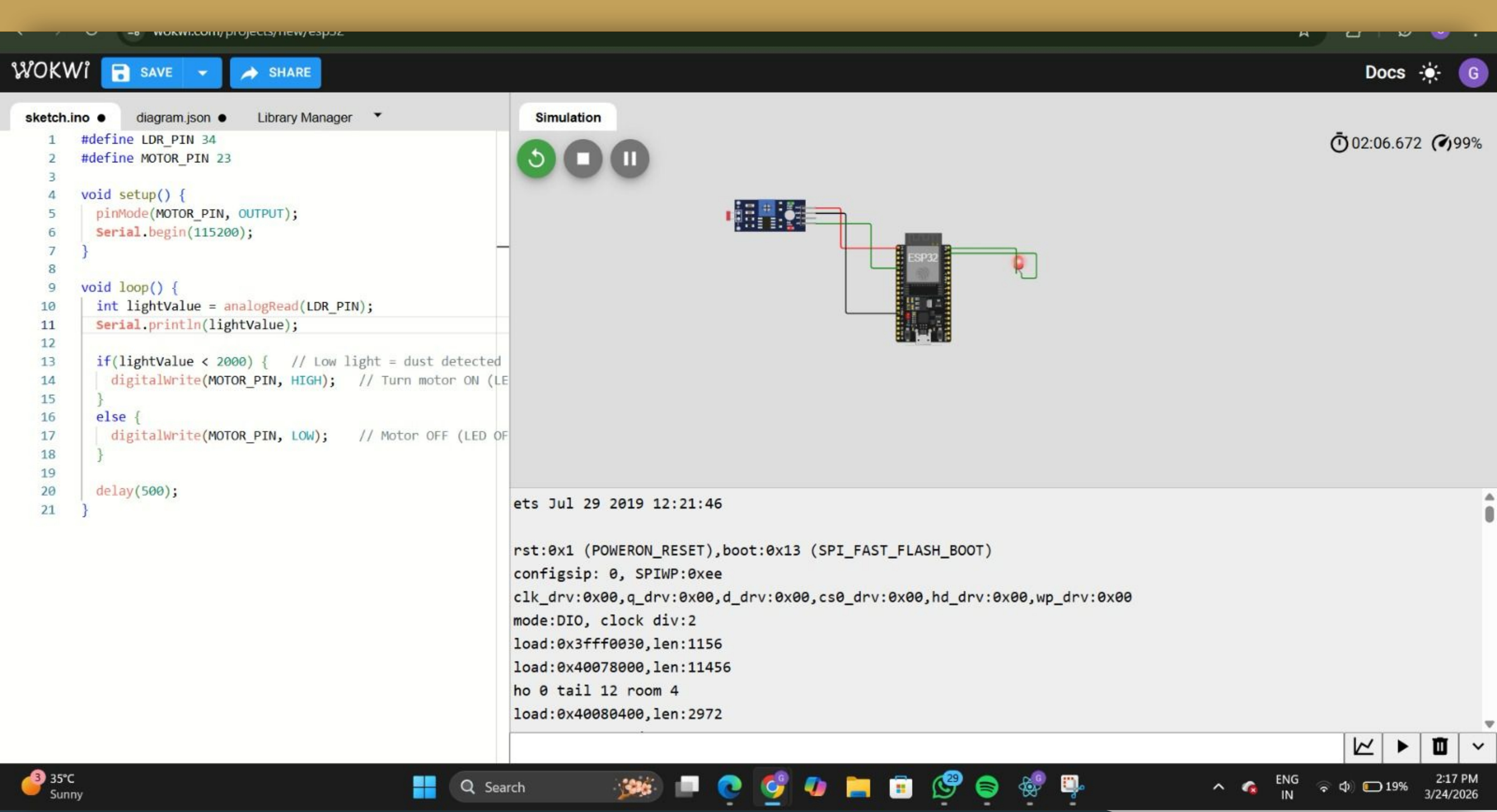
- ESP32 / NodeMCU
- LDR or Dust Sensor
- DC Motor
- Motor Driver (L298N)
- Solar Panel (Mini for demo)
- Power Supply / Battery
- Connecting Frame + Wiper

Software Components

- Arduino IDE
- Embedded :C/ C++
- IoT Platform:
  - Blynk
- Mobile dashboard for monitoring

Working Principle

- LDR compares light intensity before and after dust accumulation.
- If threshold value drops: → ESP32 activates motor.
- Motor rotates brush across panel.
- Cleaning completes.
- Data uploaded to IoT platform.


WOKWI
SAVE
SHARE
Docs
sketch.ino
diagram.json
Library Manager
#define LDR_PIN 34
#define MOTOR_PIN 23
void setup() {
pinMode(MOTOR_PIN, OUTPUT);
Serial.begin(115200);
}
void loop() {
int lightValue = analogRead(LDR_PIN);
Serial.println(lightValue);
if(lightValue < 2000) { // Low light = dust detected
digitalWrite(MOTOR_PIN, HIGH); // Turn motor ON (LE
}
else {
digitalWrite(MOTOR_PIN, LOW); // Motor OFF (LED OF
}
delay(500);
}
Simulation
02:06.672
99%
ESP32
ets Jul 29 2019 12:21:46
rst:0x1 (POWERON_RESET),boot:0x13 (SPI_FAST_FLASH_BOOT)
configsip: 0, SPIWP:0xee
clk_drv:0x00,q_drv:0x00,d_drv:0x00,cs0_drv:0x00,hd_drv:0x00,wp_drv:0x00
mode:DIO, clock div:2
load:0x3fff0030,len:1156
load:0x40078000,len:11456
ho 0 tail 12 room 4
load:0x40080400,len:2972
35°C
Sunny
Search
ENG
IN
19%
2:17 PM
3/24/2026

CODE

```
// ------------------ SETUP ------------------
void setup() {
  Serial.begin(115200);

  pinMode(relayPin, OUTPUT);
  digitalWrite(relayPin, HIGH);

  WiFi.begin(ssid, password);

  Serial.print("Connecting...");
  while (WiFi.status() != WL_CONNECTED) {
    delay(500);
    Serial.print(".");
  }

  Serial.println("\nConnected!");
  Serial.println(WiFi.localIP());

  server.on("/", handleRoot);
  server.on("/on", handleOn);
  server.on("/off", handleOff);

  server.begin();
}

// ------------------ LOOP ------------------
void loop() {
  server.handleClient();

  int ldrValue = analogRead(ldrPin);
  Serial.println(ldrValue);

  // AUTO MODE (LDR)
  if (ldrValue < threshold && !pumpState) {
    Serial.println("Dust detected → AUTO CLEAN");
    setPump(true);
  }

  // Auto OFF timer
  if (pumpState && millis() - pumpStartTime > autoOffTime) {
    setPump(false);
  }

  delay(500);
}
```

```
#include <WiFi.h>
#include <WebServer.h>

const char* ssid = "Harshitha";
const char* password = "harshi123";

WebServer server(80);

// Pins
const int relayPin = 23;
const int ldrPin = 34;

// Variables
bool pumpState = false;
unsigned long autoOffTime = 10000;
unsigned long pumpStartTime = 0;

// LDR threshold (adjust this)
int threshold = 2000;

// ---------------- RELAY ----------------
void setPump(bool state) {
  pumpState = state;

  if (state) {
    digitalWrite(relayPin, LOW);
    pumpStartTime = millis();
    Serial.println("Pump ON");
  } else {
    digitalWrite(relayPin, HIGH);
    Serial.println("Pump OFF");
  }
}

// ---------------- WEB ----------------
void handleRoot() {
  server.send(200, "text/html",
    "<h1>Solar Cleaning</h1><a href=\"/on\">ON</a><br><a href=\"/off\">OFF</a>");
}

void handleOn() {
  setPump(true);
  server.send(200, "text/plain", "ON");
}
```

Advantages:

- Automatic cleaning
- Reduces human effort
- Improves solar efficiency
- Real-time monitoring
- Scalable solution

Applications

- Residential rooftops
- Solar farms
- Industrial buildings
- Smart cities
- Remote installations

Conclusion

- Dust reduces efficiency significantly.
- Proposed IoT system automates cleaning.
- Reduces cost and maintenance effort.
- Scalable and practical smart energy solution.

References:

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